3.24 Fabric Manufacturing

Course code: **FE 253-0723**
Course type: **Core**
Total contact hours: **42**

Level/Term: **L2/T1**
Credits: **3**
CIE:TEE marks: **40:60**

Rationale

To provide required knowledge about the fundamentals of fabric manufacturing and its application in the field of Textiles. Also, students are expected to learn the basics and principles of fabric manufacturing machines, quality control instruments, and the processes of using those instruments.

Learning Outcomes

After completion of the course, the students will be able to:

CLO1	Explain the basics of all fabric production technologies and techniques
CLO2	Compare the machinery used to produce different types of fabric
CLO3	Support the onsite production and laboratory quality control systems and related machinery

Mapping of Course Learning Outcomes to Program Learning Outcomes

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1									3			
CLO2									2			
CLO3										3		

Course Contents

- **Weaving:** Introduction, Flow chart of weaving; Weaving preparation: Types of winding package, Winding machines; Warping: Definition and classification of warping, warping machines; Sizing: Size materials and their function, typical recipes; Looming: Introduction with drafting, drawing, denting, pinning, tying- in and gaiting; Weaving: Classifications of looms and description of various types of conventional looms; Picking: definition, Classification; Beat-up: definition, Classification; Weaving issues: Features of a modern loom. Brief study on air jet looms.

- **Knitting:** Introduction, General terms and principles of knitting technology. Knitting action of latch, bearded and compound needle. Elements of knitted loop structure; Basic weft knitted structures: Characteristics, Notations, Identification and uses; Basic loop or stitch type: Definition and knitting action of Tuck, Held and float stitches; Weft knitting machine: Classification, plain circular latch needle machine- Description, knitting system. Circular Rib machine and Flat knitting machine: Description.

- **Warp knitting:** Introduction and basic elements of warp knitting.

Lesson Plan

Week	Course contents	CLO
	Part A: Weaving	
1	Introduction, Flow chart of weaving; Weaving preparation:	CLO2
2	Types of winding packages, Winding machines	CLO3
3	Warping: Definition and classification of warping, warping machines;	CLO1 CLO2

4	Sizing: Size materials and their function, typical recipes; Looming:	CLO3
5	Introduction with drafting, drawing, denting, pinning, tying- in, and gaiting; Weaving:	CLO1 CLO2
6	Classifications of looms and description of various types of conventional looms; Picking: definition, Classification; Beat-up: definition, Classification;	CLO3
7	Weaving issues: Features of a modern loom. Brief study on air jet looms.	CLO1 CLO2
8	**Co-curricular activities and Sports Week**	
	Part B: Knitting	
9	Introduction, General terms and principles of knitting technology.	CLO1
10	Knitting action of latch, bearded and compound needle. Elements of knitted loop structure	CLO3
11	Basic weft knitted structures: Characteristics, Notations, Identification and uses; Basic loop or stitch type	CLO1 CLO2 CLO3
12	Definition and knitting action of Tuck, Held and float stitches	CLO1
13	Weft knitting machine: Classification, plain circular latch needle machine	CLO1
14	Description, knitting system. Circular Rib machine and Flat knitting machine	CLO3
15	Warp knitting: Introduction and basic elements of warp knitting.	CLO1
15	**Total**	**42 hrs**

Resources

- Adanur, Sabit. Handbook of weaving. CRC Press, 2000.
- Lord, Peter R., and Mansour H. Mohamed. Weaving: Conversion of yarn to fabric. Elsevier, 1982.
- Spencer, David J. Knitting technology: a comprehensive handbook and practical guide. Vol. 16. CRC Press, 2001.
- Au, Kin-Fan, ed. Advances in knitting technology. Elsevier, 2011.

Teaching-Learning Strategy: According to the approved academic rules.
Assessment Strategy: According to the approved academic rules.
Assessment Plan

Bloom's Category	CIE* (40 marks)			TEE** (60 marks)
	Class Performance (10)	Class Test (20)	Assignment (10)	
Remember				10%-20%
Understand	10	10		30%-50%
Apply		10		30%-40%
Analyse			10	
Evaluate				
Create				

*CIE- Continuous Internal Evaluation **TEE- Semester End Examination

3.24 Fabric manufacturing Lab

Course code: **FE 254-0723**
Course type: **Core**
Total contact hours: **28**

Level/Term: **L2/T1**
Credits: **1**
CIE:TEE marks: **60:40**

Learning Outcomes

After completion of the course, the students will be able to:

CLO1	Distinguish different production steps of fabric manufacturing
CLO2	Set the machine parameters based on the expected fabric types.
CLO3	Recommend a fabric production technique with the necessary calculation